

Database Project Proposal Form

Project

Project Title:	<i>Hotel Reservation Management System</i>	
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Statement of Problem:	<i>Most hotels currently rely on manual or fragmented systems to manage room reservations, guest check-ins, and payments. This leads to overbooking errors, data redundancy, slow retrieval of booking history, and poor coordination between departments.</i> <i>A centralized database system is needed to automate and streamline all hotel operation</i>	
Brief Description of the Project:	<i>The Hotel Reservation Management System is a relational database system designed to manage all aspects of hotel operations. It handles room availability, customer bookings, check-in/check-out processes, payment records, and staff management. Users include hotel administrators, receptionists, and guests. The system ensures real-time room availability tracking, billing accuracy, and complete booking history for better decision-making and guest satisfaction.</i>	
Objectives:	• <i>To Organize and centralize all hotel data efficiently.</i> • <i>To Eliminate data redundancy and manual booking errors.</i> • <i>To Enable fast retrieval of room availability and guest records.</i> • <i>To Automate billing and payment processing.</i> • <i>To Support role-based access for admins, receptionists, and guests.</i>	

Technical Approach:	• <i>DBMS: MYSQL/SQL Server</i> • <i>Entity-Relationship Diagram (ERD) to model all entities and relationships.</i> • <i>Normalized relational tables: Rooms, Customers, Reservations, Payments, Staff.</i> • <i>SQL queries: SELECT, INSERT, UPDATE, DELETE, JOINS, and aggregate functions.</i> • <i>Stored procedures and views for common operations.</i>
Expected Deliverables:	• <i>Complete and normalized database schema.</i> • <i>Entity-Relationship Diagram (ERD).</i> • <i>SQL scripts (DDL + DML).</i> • <i>Full project documentation and report</i> • <i>Desktop App (Bonus Task)</i>